WORKED SOLUTION

STEP III 1988, Question 12

Mechanics • Collision

STEP III 1988, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/88-s3-q12

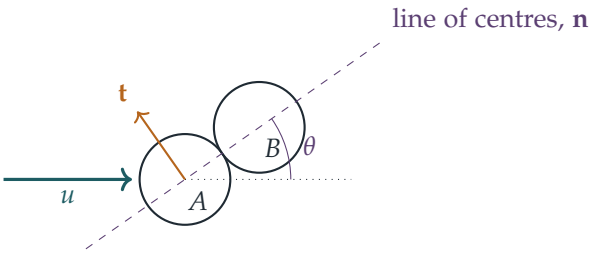
The key idea

For smooth spheres the impulse acts only along the line of centres at the moment of impact. So we split every velocity into a component along that line and one perpendicular to it: the perpendicular components are untouched, and along the line we have an ordinary one-dimensional collision. Everything then depends on a single unknown, the angle between the line of centres and the first ball's original velocity.

Solution

Setting up

Model the balls as smooth spheres of equal mass m, sliding on a smooth table, so the only horizontal force during impact is the contact impulse along the line of centres. Let the first ball A move with speed u, and let the line of centres make angle θ with A's velocity at impact, where $0 \le \theta < \frac{1}{2}\pi$ (if $\theta = \frac{1}{2}\pi$ the balls only graze and nothing happens). Take unit vectors $\mathbf{n}$ along the line of centres (from A to B) and $\mathbf{t}$ perpendicular to it.



Before impact A has components $u\cos\theta$ along $\mathbf{n}$ and $u\sin\theta$ along $\mathbf{t}$; B is at rest.

Along $\mathbf{t}$: no impulse acts in this direction on either ball, so A keeps $u\sin\theta$ and B keeps 0.

Along $\mathbf{n}$*:* let the components afterwards be v_A and v_B. Conservation of momentum (the masses cancel) and Newton's law of restitution give

$$v_A + v_B = u\cos\theta, \qquad v_B - v_A = e\,u\cos\theta,$$

so

$$v_A = \tfrac{1}{2}(1-e)\,u\cos\theta, \qquad v_B = \tfrac{1}{2}(1+e)\,u\cos\theta.$$

So afterwards

$$\mathbf{v}_A = \tfrac{1}{2}(1-e)u\cos\theta\,\mathbf{n} + u\sin\theta\,\mathbf{t}, \qquad \mathbf{v}_B = \tfrac{1}{2}(1+e)u\cos\theta\,\mathbf{n}.$$

The angle between the velocities

B moves along the line of centres, and the angle between the two velocities is acute exactly when their scalar product is positive. Here

$$\mathbf{v}_A \cdot \mathbf{v}_B = \tfrac{1}{4}(1-e)(1+e)\,u^2\cos^2\theta > 0,$$

because $e < 1$, $u \neq 0$ and $\cos\theta > 0$ for $0 \le \theta < \tfrac{1}{2}\pi$. Hence the angle between the velocities is less than $\tfrac{1}{2}\pi$, as required. (For a head-on impact, $\theta = 0$, both balls move along the same line and the angle is 0.)

Note where $e < 1$ was used: if $e = 1$ then $v_A = 0$ and the balls separate at exactly a right angle, the familiar snooker result.

The maximum deflection

Let ϕ be the angle through which A is deflected, i.e. the angle between $\mathbf{v}_A$ and the original direction $\cos\theta\,\mathbf{n} + \sin\theta\,\mathbf{t}$. Write $k = \tfrac{1}{2}(1-e)$, so $0 < k \le \tfrac{1}{2}$ and $\mathbf{v}_A = u(k\cos\theta\,\mathbf{n} + \sin\theta\,\mathbf{t})$. Using the scalar product and the two-dimensional cross product of the direction vectors $(\cos\theta, \sin\theta)$ and $(k\cos\theta, \sin\theta)$,

$$|\mathbf{v}_A|\cos\phi = u\,(k\cos^2\theta + \sin^2\theta),$$
$$|\mathbf{v}_A|\sin\phi = u\,(\cos\theta\sin\theta - k\cos\theta\sin\theta) = u\,(1-k)\sin\theta\cos\theta.$$

The first is positive, so ϕ is acute, and dividing (then dividing top and bottom by $\cos^2\theta$) with $t = \tan\theta \ge 0$,

$$\tan\phi = \frac{(1-k)t}{k+t^2}.$$

Ask yourself How do we maximise this without a messy derivative?

Since ϕ is acute, maximising ϕ is the same as maximising $\tan\phi$. For $t > 0$,

$$\tan\phi = \frac{1-k}{k/t + t} \quad \text{and} \quad \frac{k}{t} + t \ge 2\sqrt{k}$$

by the AM–GM inequality, with equality exactly when $t = \sqrt{k}$. (At $t = 0$, $\phi = 0$.) So the maximum is attained when $\tan\theta = \sqrt{k}$, and

$$\tan\phi_{\max} = \frac{1-k}{2\sqrt{k}}.$$

To convert to a sine, note $(1-k)^2 + (2\sqrt{k})^2 = 1 - 2k + k^2 + 4k = (1+k)^2$, so in the right-angled triangle with legs $1-k$ and $2\sqrt{k}$ the hypotenuse is $1+k$, and

$$\sin\phi_{\max} = \frac{1-k}{1+k} = \frac{\frac{1}{2}(1+e)}{\frac{1}{2}(3-e)} = \frac{1+e}{3-e},$$

using $1-k = \frac{1}{2}(1+e)$ and $1+k = \frac{1}{2}(3-e)$. Since $\phi_{\max}$ is acute,

$$\phi_{\max} = \sin^{-1}\left(\frac{1+e}{3-e}\right),$$

which is the required result. It occurs when the line of centres makes angle $\tan^{-1}\sqrt{\frac{1}{2}(1-e)}$ with the original motion.

Checks

- $e \to 1$: the formula gives $\sin^{-1} 1 = \frac{1}{2}\pi$, consistent with the perfectly elastic case: there A loses all its velocity along the line of centres, so it always moves at right angles to that line, and a nearly head-on blow sends it off at nearly a right angle to its original path.
- $e = 0$: $\sin\phi_{\max} = \frac{1}{3}$, so $\phi_{\max} \approx 0.3398$ rad (19.5°).
- $e = \frac{1}{2}$: $\sin\phi_{\max} = \frac{3}{5}$, $\phi_{\max} \approx 0.6435$ rad, at $\theta = \tan^{-1}\frac{1}{2} \approx 0.4636$. A numerical scan of the deflection over 200 000 values of θ reproduced both numbers, and 200 000 random collisions never gave an angle between the velocities of $\frac{1}{2}\pi$ or more.

Common slips

- Applying restitution to the full speeds rather than to the components along the line of centres.
- Forgetting that the perpendicular component of A's velocity is unchanged; this is what makes A deflect at all.
- Maximising ϕ itself by differentiating an inverse tangent; working with $\tan\phi$ (legitimate because ϕ is acute) is far cleaner.

Beyond STEP

The decomposition here is the impulse form of Newton's second law: impulses change momentum only in the direction they act. In the elastic case the right-angle result is a special case of a neat frame-of-reference argument: in the centre-of-mass frame an elastic collision keeps each ball's speed and only turns its direction, and the transformation back to the lab frame gives every possible outcome at once. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats collisions in the centre-of-mass frame this way.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.